Ashwin Balakrishna

Frontend Architect | React.js | TypeScript | Micro Frontend Architecture | AI-Native Engineering

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PROFESSIONAL SUMMARY

Frontend Architect with 5+ years owning end-to-end architectural decisions, team leadership, and delivery across enterprise and startup products. Deep expertise in React.js, TypeScript, Next.js, and Micro Frontend architecture using Webpack Module Federation — currently spearheading the migration to Vite-based MFE with Zustand as the shared state backbone. Delivered the zero-regression react-router-dom v5 to v7 upgrade, a backlog item deferred for nearly two years. Active contributor to product roadmap and feature design on a real-time BPO-grade two-way chat platform replacing lock-based multi-operator workflows. AI-native practitioner using Claude Code, GitHub Copilot, and Codex. Measurable impact: 63%+ page load improvement, 90%+ test coverage enforced as CI/CD gates, and full ownership of architectural evolution across production systems.

KEY ACHIEVEMENTS

- **Vite + Module Federation Migration:** Architecting and leading the transition from Webpack-based MFE to Vite-powered Module Federation with Zustand as the shared global state layer — improving dev build times significantly and modernising the entire frontend platform.
- **React Router v5 to v7 (2-year backlog):** Delivered zero-regression upgrade across a live MFE/PWA codebase, unblocking modern routing APIs and closing a long-standing architectural debt item deferred for nearly two years.
- **BPO Chat Platform Architecture:** Designed and drove feature architecture for a real-time two-way chat application for BPO call centre operations — eliminating lock-based contention and enabling multiple operators to manage multiple concurrent conversation topics with no queue collisions.
- **AWS Parameter Store Integration:** Implemented Parameter Store-based env config access via window-level globals, decoupling environment-specific values from build artifacts and enabling dynamic, secure runtime config across all MFE modules.
- **LaunchDarkly Full Adoption:** Led LaunchDarkly from POC to production — enabling phased rollouts and A/B testing across the React platform, eliminating risky big-bang releases.
- **63.3% Page Load Improvement** on 2Hub via Webpack optimisation, lazy loading, and Core Web Vitals best practices. | **90%+ Test Coverage** maintained across all projects with Jest + RTL, enforced as mandatory CI/CD quality gates.

PROFESSIONAL EXPERIENCE

Frontend Architect / Software Engineer I **Payoda Technologies** Jun 2024 – Present

- **Architecture Ownership:** Own all frontend architectural decisions — including MFE topology, shared dependency strategy, state management selection, and package upgrade roadmap. Primary decision-maker for when and how architectural evolution is introduced into the production codebase.
- **Vite + MFE Migration:** Currently leading migration of Webpack-based MFE architecture to Vite-powered Module Federation, improving DX and build performance. Evaluating and implementing Zustand as the unified shared state layer across all MFE modules.
- **BPO Chat — Product and Architecture:** Actively driving roadmap decisions and feature architecture for a two-way real-time chat product serving BPO call centre workflows. Designed the solution to eliminate lock-based operator-chat contention — enabling multiple operators to handle concurrent multi-topic threads without queue collisions.
- **AWS Parameter Store:** Implemented secure runtime environment config via AWS Parameter Store, surfacing values through window-level globals — decoupling environment state from build artifacts across all MFE modules.
- **Router Migration:** Executed zero-regression react-router-dom v5 to v7 across a live MFE/PWA codebase — a migration deferred in the backlog for nearly two years — unblocking modern routing patterns and nested route APIs for the entire frontend team.
- **LaunchDarkly:** Drove LaunchDarkly from POC to full production, enabling phased rollouts and A/B testing across the platform.

- **AI-Native Engineering:** Embedded Claude Code, GitHub Copilot, and Codex into daily engineering workflow — improving velocity on boilerplate generation, refactoring, and code review.
- **Quality Engineering:** Maintained 90%+ Jest + React Testing Library coverage; enforced as mandatory CI/CD quality gates on all pull requests.

Senior Product Engineer Codingmart Technologies Mar 2021 – Jun 2024

- **Aditya Birla Fashion and Retail — Tech Lead:** Led a team of 3 frontend engineers to build the CMS Dashboard from scratch. Owned architecture decisions including Next.js SSR/SSG strategy, Redux Toolkit state design, REST API integration patterns, and component system structure. Delivered on schedule with zero critical bugs at launch.
- **Performance Engineering:** Achieved 63.3% improvement in page load time on the 2Hub travel booking platform via Webpack bundle optimisation, lazy loading, and debouncing — directly improving Core Web Vitals.
- **SEO and Accessibility:** Improved SEO rankings and WCAG a11y compliance across 3 client projects using semantic HTML, structured data, and systematic accessibility audits.
- **CI/CD Design and Mentorship:** Designed end-to-end CI/CD pipelines; mentored junior devs on React/TypeScript best practices; enforced ESLint/Prettier standards team-wide.

Founder and Lead Frontend Engineer Aztrikz Tech Aug 2019 – Mar 2021

- Founded a web solutions startup; delivered 3 end-to-end client products using React.js, Next.js, and WordPress — managing architecture, development, CI/CD pipelines, and AWS deployments.

KEY PROJECTS

Solutions by Text — BPO Two-Way Chat Platform (MFE PWA) — Payoda Technologies

- Architected and own 10+ MFE modules in production using React.js, TypeScript, and Webpack Module Federation; currently leading Vite-based MFE migration with Zustand as the shared global state layer.
- Designed real-time two-way chat architecture for BPO call centre operations — eliminating lock-based multi-operator contention and enabling concurrent multi-topic handling per operator.
- Implemented AWS Parameter Store integration for window-level runtime env config across all MFE modules. Delivered zero-regression react-router-dom v5 to v7 migration — a 2-year deferred backlog item — with no downtime.

Aditya Birla Fashion and Retail — CMS Dashboard — Codingmart Technologies

- Tech Lead for a team of 3; owned architecture, sprint planning, and delivery of a greenfield CMS dashboard using Next.js, Redux Toolkit, SSR/SSG, TypeScript, and REST APIs.
- Drove HOC patterns and Redux Toolkit optimisations to reduce bundle size; enforced responsive design and WCAG a11y standards. Delivered on schedule with zero critical bugs at launch.

2Hub — Hotel and Travel Booking Platform — Codingmart Technologies

- Achieved 63.3% page load improvement via Webpack bundle optimisation — measurably improving Core Web Vitals and user retention.
- Engineered automated CI/CD pipeline, reducing deployment time and eliminating manual release errors entirely.

TECHNICAL SKILLS

Frontend: React.js, TypeScript, JavaScript (ES6+), Next.js, Tailwind CSS, HTML5, CSS3

State Management: Redux Toolkit, Zustand, React Query, Context API

Architecture: Micro Frontend (MFE), Webpack Module Federation, Vite MFE (in progress), SSR/SSG, PWA, WCAG/a11y, Core Web Vitals

Testing: Jest, React Testing Library (RTL), TDD, 90%+ coverage

Cloud and DevOps: AWS (Parameter Store, deployments), Webpack, Vite, CI/CD pipelines, Docker

AI-Native: Claude Code, GitHub Copilot, Codex, Cursor | **Tools:** LaunchDarkly, Storybook, Git, Agile/Scrum

EDUCATION AND CERTIFICATIONS

Bachelor of Engineering, Computer Science and Engineering — Anna University, India (2021)

Generative AI for Software Development · The Complete AI Coding Course (2025) — Cursor, Claude Code